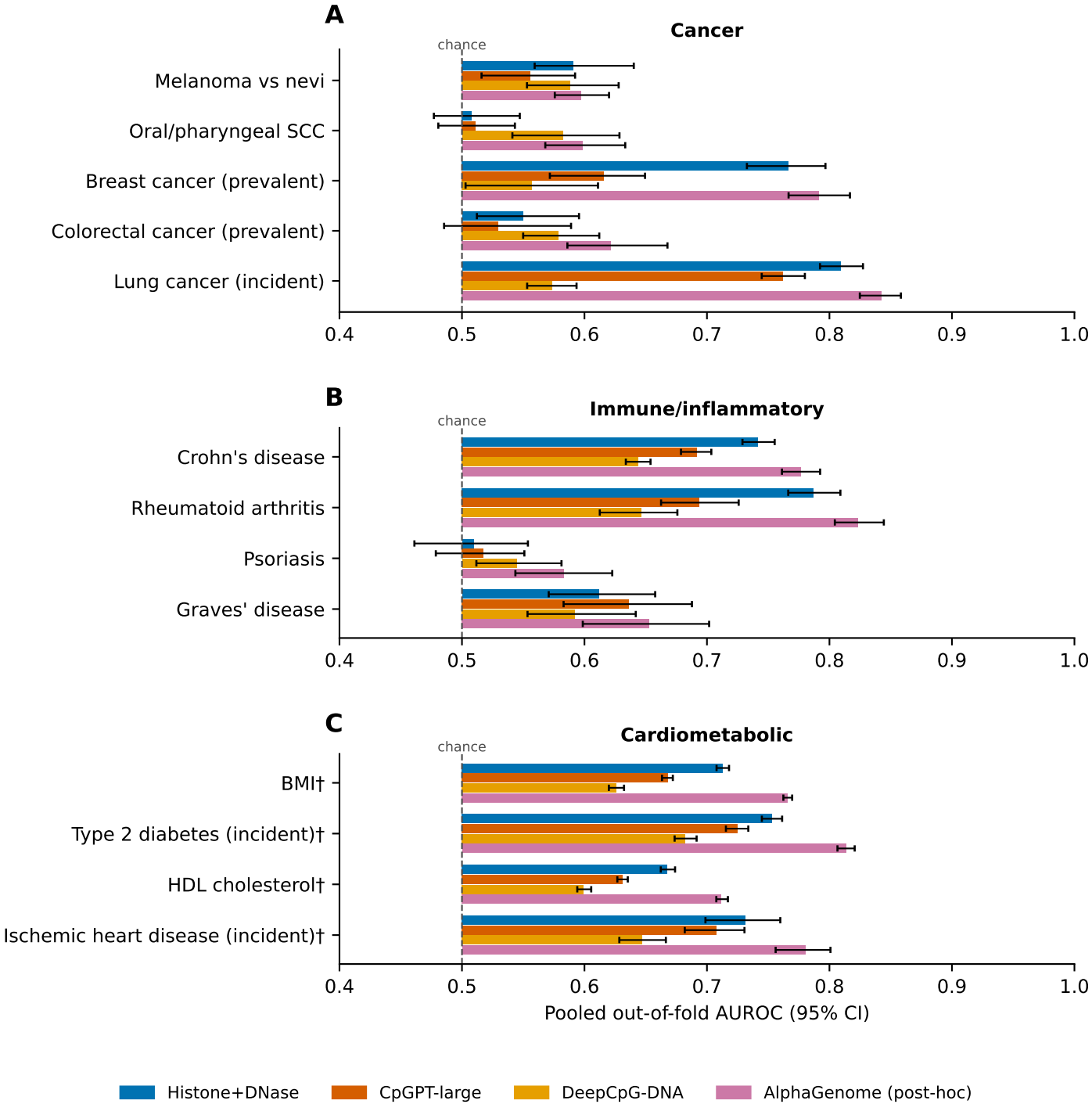

EWAS Catalog-associated CpGs: $p < 10^{-4}$; fixed 408,399-CpG universe.

† >50% overlap with another selected task; shared cohorts are not independent replications.

AlphaGenome: post-hoc comparator (posthoc_comparator: true), added after the original-arm results; same frozen pairs, probe and bootstrap draws; outside the approved 3-arm panel and every inferential family.